

An Assessment of DNSSEC Amplification in 2026

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Abstract

DNS amplification attacks exploit the disproportion between small queries and large responses to overwhelm a victim. DNSSEC introduces large cryptographic records that increase this disproportion. We present a large-scale active measurement study of DNSSEC amplification across the Tranco top-1M domains, finding that only 6.7% of domains are fully signed. Signed domains produce substantially higher amplification than unsigned ones, with RRSIG queries posing the greatest risk. We further find that RSA-based algorithms produce significantly higher amplification than elliptic curve alternatives for both DNSKEY and RRSIG queries, suggesting that the ongoing migration toward elliptic curve algorithms can reduce DNSSEC amplification risk.

1 Introduction

The Domain Name System (DNS) is a core component of the internet, used to translate between human readable domain names and IP addresses. DNS was not designed with security in mind, making it vulnerable to a range of attacks. Of these, amplification attacks are notably concerning: An attacker sends DNS queries with a spoofed source address of a victim, directing large responses from small queries [1].

DNS Security Extensions (DNSSEC) were introduced to add authenticity and integrity to DNS [2]. However, the cryptographic records that DNSSEC adds to zones: public keys, signatures, and denial-of-existence records are larger than standard DNS records, raising the concern that DNSSEC may extend the amplification risk. Prior work has confirmed that DNSSEC enabled queries produce higher amplification factors than standard DNS queries [3]. Since these studies were conducted, DNSSEC deployment has likely increased, and changes in algorithm usage, key management, and zone

configurations may have changed the amplification landscape.

This paper revisits DNSSEC amplification with a large-scale active measurement study of the Tranco top-1M domains, issuing DNSKEY, DS, RRSIG, and NSEC queries and characterizing the amplification behavior of both signed and unsigned domains. We address the following research questions:

- **RQ1:** What is the distribution of DNSSEC signed and unsigned domains among the top 1M?
- **RQ2:** What is the distribution of amplification factors for DNSSEC enabled queries?
- **RQ3:** How do amplification factors differ between signed vs unsigned domains?
- **RQ4:** How do DNSSEC properties (i.e. key size, algorithm) affect amplification?

2 Background

2.1 DNS and Amplification Attacks

The Domain Name System operates as a hierarchical, distributed database resolved through a chain of authoritative nameservers [1]. Clients query recursive resolvers, which traverse the hierarchy on their behalf. DNS is frequently used over UDP, allowing an attacker to spoof the source address of a query, directing the resolver's response to a victim. The amplification factor represents the degree to which an attacker can magnify their traffic volume. High amplification factors indicate a threat, as they allow an attacker with limited bandwidth to generate a disproportionately large attack.

2.2 DNSSEC

DNSSEC is a set of extensions to DNS standardized by the IETF in 2005 [4; 5; 6] that provides authentication of DNS data, authenticated denial of existence, and data integrity. Zone operators cryptographically sign their zones using public key cryptography, and validating resolvers can verify these signatures against published pub-

lic keys. DNSSEC introduces several resource record types that are central to this paper:

- **DNSKEY:** Publishes the public keys used to sign a zone. A zone typically maintains a Key Signing Key (KSK) that signs the **DNSKEY** resource record set (RRset) itself, and a Zone Signing Key (ZSK) that signs all other records in the zone.
- **DS:** A Delegation Signer record that is held in the parent zone. It contains a hash of the child zone’s KSK and establishes the chain of trust from a resolver to domain.
- **RRSIG:** A signature over an RRset. Every signed RRset in a zone has a corresponding **RRSIG**, so zones with many record types produce many **RRSIG** records per response.
- **NSEC / NSEC3:** Next Secure records, used to provide authenticated denial of existence. An **NSEC** links to the next record name in the zone to prove non existence of a record. **NSEC3** uses hashed versions of names to prevent the Zone Walking vulnerability [7].

Each of these record types adds to the size of DNS responses, adding data that creates the amplification potential examined in this paper.

3 Methodology

We conduct active DNS measurements over the Tranco top-1M list [8], a widely used ranking in the research community. We use the version published on February 2nd, 2026. For each domain, we issue **DNSKEY**, **DS**, **RRSIG**, and **NSEC** queries using **dig** with EDNS0 enabled and a buffer size of 4096 bytes, directed at Google’s public resolver (8.8.8.8). By using an accessible public resolver, we can mimic how a real world attacker would exploit such infrastructure. We compute the amplification factor for each response as:

$$\text{Amplification Factor} = \frac{\text{Response size (bytes)}}{\text{Query size (bytes)}} \quad (1)$$

Measurements were automated via a Python script and executed from an Ubuntu 20.04 virtual machine at CWRU. Preliminary measurements were conducted with variations in query types, resolvers, and subset sizes to calibrate data collection and processing scripts. The dataset used in this paper was collected over March 28–April 7, 2026.

Limitations All queries originate from a single vantage point and resolver. This may introduce bias caused by resolver specific caching or implementation differences. Additionally, our dataset is limited to the Tranco

top-1M, which biases towards popular, well maintained domains. While we found that such domains exhibit the behavior this study focuses on, this dataset may not represent DNSSEC trends in the broader domain landscape.

Query Density We issue **DNSKEY** and **DS** queries to all 1 000 000 domains in the dataset, which established the rank-dependent signing trend described in Section 4.1. Since signed domains are the primary focus of the remaining analysis, we adopt a stratified sampling method for **RRSIG** and **NSEC** queries, querying domains at decreasing density as Tranco rank increases:

- Ranks 1–100k: every domain
- Ranks 100k–250k: every 2nd domain
- Ranks 250k–500k: every 5th domain
- Ranks 500k–750k: every 10th domain
- Ranks 750k–1M: every 20th domain

To validate this approach, we compute a weighted signing rate from the sampled dataset, where each signed domain as defined in Section 4.1 is scaled by its respective sampling interval. The resulting weighted signing rate of 6.7% matches the rate observed from the full 1 000 000-domain **DNSKEY** and **DS** measurement, confirming that the stratified sample is representative of the broader population.

Ethics All measurements were conducted in a manner designed to minimize potential harm and avoid disruption to external systems. Queries were issued at a controlled rate to prevent overloading resolvers or authoritative servers, and no attempts were made to exploit amplification against third-party victims. All queries were sent from and responses received by our own measurement infrastructure, so any amplification effects were effectively self-directed.

4 Results

4.1 RQ 1: Deployment of DNSSEC

Understanding the landscape of DNSSEC deployment is an important first step in studying the amplification behaviors of DNSSEC enabled queries. Prior work has measured DNSSEC adoption through active probing, querying authoritative nameservers to determine whether zones are signed [9; 1]. Following this methodology, we apply a three way classification scheme to the domains in our dataset.

A domain’s signing status is determined by issuing both **DNSKEY** and **DS** queries and observing whether non-empty answers are returned. **DNSKEY** records publish the public keys used to sign a zone, while **DS** records establish the chain of trust required for DNSSEC validation [5;

6]. Non empty responses to both queries indicates that a domain is signed and correctly delegated. We define three categories:

- **Signed:** the domain returns a non-empty answer to both DNSKEY and DS queries, indicating a fully deployed and delegated DNSSEC chain of trust.
- **Unsigned:** the domain returns no answers for either DNSKEY or DS queries, indicating the zone has not been signed.
- **Misconfigured:** the domain returns a non-empty answer for only one of the two queries. This may indicate an incomplete deployment, such as a zone that has been signed locally but whose DS record has not yet been published by the parent, or a DS record whose corresponding DNSKEY has been removed.

Full Dataset Table 1 presents the distribution of DNSKEY and DS responses across the dataset. Of the 1M domains measured, 66k (6.7%) are classified as *signed*, having returned non-empty answers to both queries. The vast majority, 902k domains (90.2%), returned no answer to either query and are therefore classified as *unsigned*. The remaining 3.1% of domains are classified as *misconfigured*: 31 053 returned a DNSKEY response without a corresponding DS record, while a much smaller subset of 402 domains exhibited the inverse, namely a DS record present at the parent with no reachable DNSKEY in the zone.

These results agree with prior work. Studies have found that DNSSEC deployment tends to be low and that a common cause is the failure to upload DS records even while DNSKEY or RRSIG records are present [3].

Category	# of Domains	Percent of Total
DNSKEY response	97703	9.8%
DS response	66926	6.7%
DNSKEY and DS response	66650	6.7%
DNSKEY response only	31053	3.1%
DS response only	402	0.0004%
Neither	902296	90.2%

Table 1: Distribution of DNSSEC responses across queried domains

By Rank The total average signing rate of 6.7% overshadows a rank dependent trend. To examine how DNSSEC adoption varies across the Tranco list, we group domains into consecutive buckets of 15 000 domains each and compute the fraction of signed domains within each bucket. Figure 1 plots this fraction against the Tranco rank for DNSKEY and DS responses. The average rate of DNSSEC adoption (6.7%) is included as a reference line. This shows a clear decrease in DNSSEC

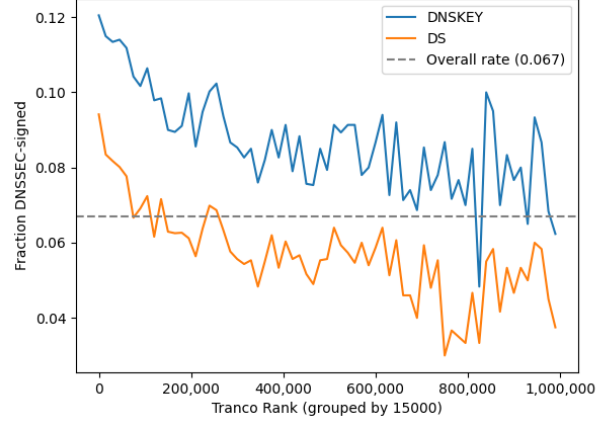


Figure 1: DNSSEC Adoption Rate by Rank

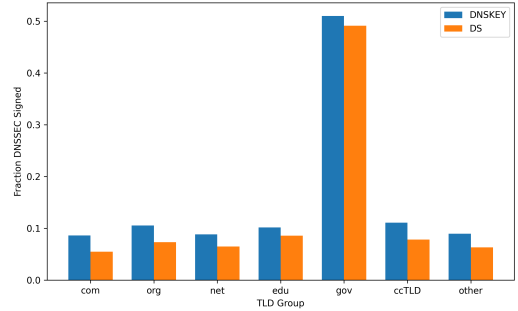


Figure 2: DNSSEC Adoption Rate by TLD

adoption as Tranco rank increases. Among the top-ranked domains, the signing rate exceeds the dataset average of 6.7%, whereas domains in the lower rank bands fall well below it.

By TLD DNSSEC adoption is mostly uniform across top-level domains (TLDs). We group domains by TLD into seven categories: the five major generic TLDs (.com, .org, .net, .edu, .gov), country-code TLDs (ccTLDs), and all remaining TLDs. Figure 2 shows the signing rates for the groups. The most notable result is the adoption rate of .gov domains, 49.1%, almost eight times the dataset average of 6.7%. This is a policy driven outlier as DNSSEC deployment was mandated for U.S. federal agencies in 2008 [10]. We note that .gov domains represent only 1.5% of the dataset, so their high signing rate has limited influence on the aggregate rate.

4.2 RQ 2: Amplification Distribution

Table 2 summarizes the amplification factor statistics for each query type across all measured domains. Across all query types, the overall median amplification factor is

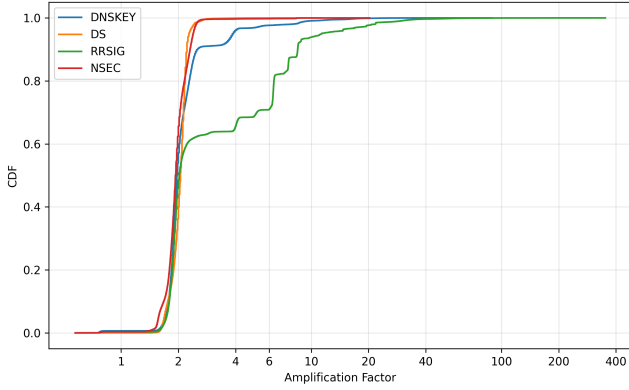


Figure 3: CDF of Amplification Factor by Query Type

2.00, indicating mostly modest amplification. However, the mean of 2.70 and a 99th percentile of 15.10 indicate right-skewed distributions with heavy tails.

The query types differ largely in their tail behavior. DS and NSEC both have medians near 2.0 and 99th percentiles below 2.7, indicating that extreme responses are rare and the distribution is tightly concentrated. DNSKEY exhibits a heavier tail, with a 99th percentile of 9.58 and a maximum of 88.83, reflecting the fact that zones with large or numerous public keys produce significantly larger responses.

Type	Mean	Median	Min	Max	P95	P99
DNSKEY	2.334	1.966	0.574	88.830	3.902	9.583
DS	2.034	2.052	0.574	11.250	2.273	2.623
RRSIG	4.475	2.020	0.574	351.588	11.352	29.151
NSEC	1.976	1.932	0.574	20.283	2.393	2.558
ALL	2.704	2.000	0.574	351.588	7.160	15.095

Table 2: Amplification factor statistics across DNSSEC record types

RRSIG exhibits the highest amplification behavior: despite a median of 2.02, its 99th percentile reaches 29.15 and its maximum of 351.59 is four times higher than any other type.

Figure 3 confirms these patterns. The DS and NSEC CDFs rise steeply and plateau early, while DNSKEY shows a limited tail. The RRSIG CDF shifts noticeably from the others past the median, extending beyond an amplification factor of 100, reflecting the high variance in the number of signed RRsets per zone.

A minimum amplification of 0.574 appears uniformly across all query types, attributable to SERVFAIL responses, which carry no answer section and thus produce a response smaller than the query. These sub-unity cases affect approximately 330 domains and are retained in the dataset for completeness but do not substantially influence the reported statistics.

5 Analysis

5.1 RQ 3: Signed vs Unsigned Behavior

By examining differences in amplification for signed vs unsigned domains, we can determine which one an attacker would prefer to target.

Table 3 compares amplification factor statistics for signed and unsigned domains collapsed across all four query types. The difference between the two groups is significant. Signed domains produce a mean amplification factor of 5.81, more than twice that of unsigned domains at 2.48.

Type	Mean	Median	Min	Max	P95	P99
UNSIGNED	2.482	1.983	0.574	191.271	6.314	10.500
SIGNED	5.809	2.474	0.745	351.588	21.537	43.035

Table 3: Amplification factor statistics for signed vs. unsigned domains across all query types.

The tail end behavior exhibits the largest difference. The 95th and 99th percentiles for signed domains reach 21.54 and 43.04 respectively, compared to 6.31 and 10.50 for unsigned domains. The maximum observed amplification factor of 351.59 belongs to a signed domain, nearly double the unsigned maximum of 191.27. This confirms that while unsigned domains can still produce noticeable amplification (such as NSEC denial-of-existence responses) the maximal amplification is concentrated among signed domains.

By Query Type Figure 4 shows the CDF of the amplification factor for each query type, split by signed and unsigned domains. For NSEC and DS queries, the signed and unsigned CDFs are very similar, with both curves rising quickly and plateauing before an amplification factor of 3. This indicates that for these records, signing has little effect on amplification. Unsigned domains return denial or referral messages that are comparably sized to answer responses.

Contrastingly, DNSKEY queries show a significant increase in amplification for signed domains. The unsigned domains follow a similar trend of low amplification and few high outliers along with NSEC and DS queries, while the signed domains have higher and more varied amplification factors. Lastly, RRSIG queries exhibit interesting behavior for both signed and unsigned domains. Unsigned RRSIG responses tend to be the largest for all unsigned query types. Then signed RRSIG responses have the largest amplification overall, rising slowly and with a long tail. RRSIG queries to signed domains carry the highest amplification risk of all query types studied — both with respect to median amplification and worst case tail end behavior.

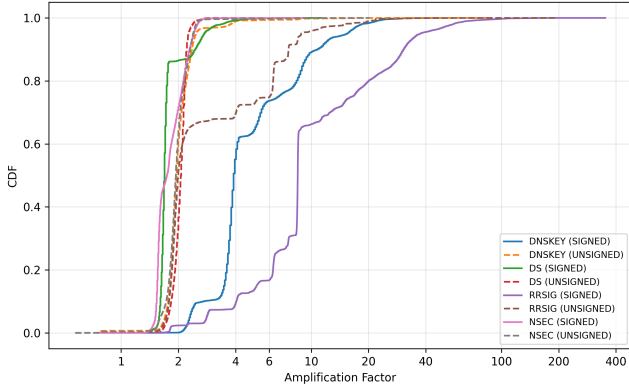


Figure 4: Amplification CDF by Query Type and DNSSEC Status

5.2 RQ 4: DNSSEC Properties

We now examine the properties of the two queries that produced the highest amplification, **DNSKEY** and **RRSIG**, to identify which DNSSEC properties are associated with high amplification.

DNSKEY

We begin by characterising the **DNSKEY** records returned for signed domains. Of the 25 403 domains that returned at least one **DNSKEY** response, 1286 returned a **CNAME** record rather than key material, indicating that these zones delegate their DNS to a third-party provider and do not host their own keys. These domains were excluded from the property analysis, leaving 24 117 domains and 51 499 individual **DNSKEY** records.

Key Count The majority of domains (72.6%) publish exactly two keys, consistent with the standard practice of maintaining a separate Key Signing Key (KSK) and Zone Signing Key (ZSK) [5; 6]. Fewer domains publish exactly 1 or 3 keys, and less than 5% of domains publish 4 or more keys. Figure 5 plots median amplification against key count. As expected, amplification increases directly with the number of keys returned: each additional **DNSKEY** record adds directly to the response size, producing growth in the median amplification factor. This confirms that key count is a primary driver of **DNSKEY** amplification.

Algorithm Figure 6 shows median **DNSKEY** amplification grouped by the primary algorithm used in each zone. ECDSA P-256 dominates the dataset, reflecting its status as the current recommended algorithm for new deployments [11]. RSA domains, while fewer in number, exhibit higher amplification. This is a consequence of RSA public keys being significantly larger than elliptic

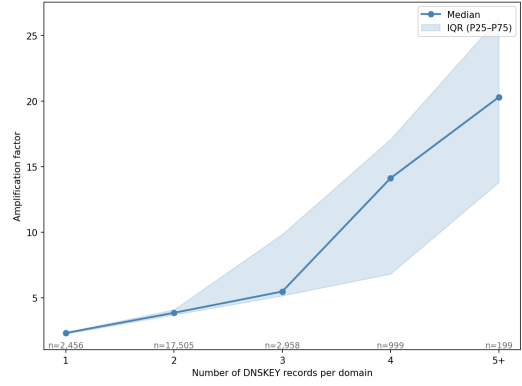


Figure 5: Median amplification vs DNSKEY key count

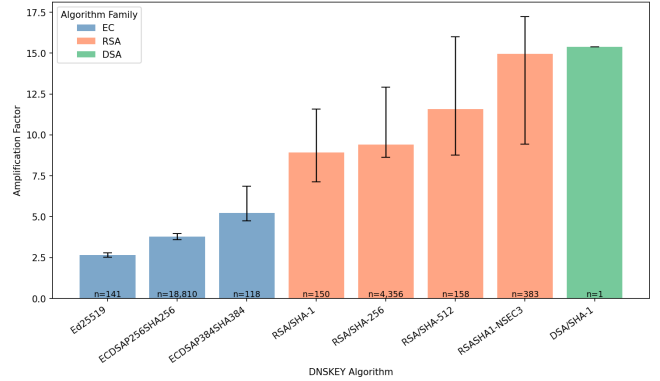


Figure 6: Median Amplification by Algorithm

curve keys: a 2048-bit RSA key occupies 256 bytes, compared to 32–64 bytes for ECDSA and Ed25519.

Key Size For RSA domains, Figure 7 examines the effect of key size on **DNSKEY** amplification. Since RSA key sizes in practice cluster around standard values, we group domains into four buckets: ≤ 1024 , 1025–2048, 2049–4096, and >4096 bits, and report the median amplification with inter quartile range for each group. The EC median is shown as a reference line ($n=19\,069$). The trend shows an increase in amplification as RSA key sizes grow across all four buckets. However, this growth is not linear with respect to key size. The step from ≤ 1024 to 1025–2048 bits produces a 70% increase in median amplification (from 3.91 to 6.83 per kilobit of key material), while the step from 1025–2048 to 2049–4096 yields a smaller relative increase of 38%. The jump to >4096 bits produces another 78% increase, reaching a median amplification of 16.7.

This pattern suggests that amplification is influenced not only by key size, but also by fixed per-record overhead in **DNSKEY** responses. At smaller key sizes, this overhead constitutes a larger fraction of the response,

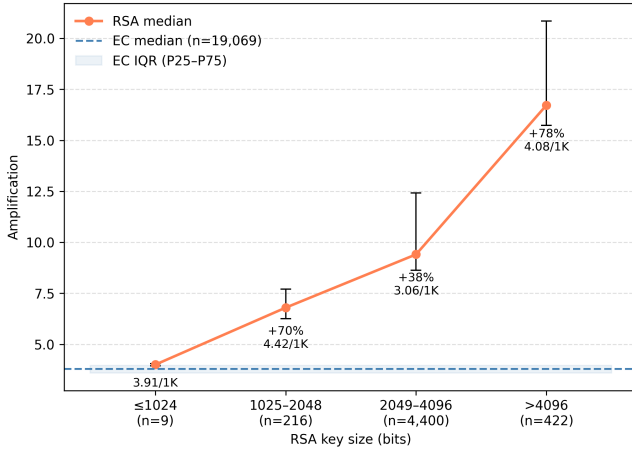


Figure 7: DNSKEY amplification vs RSA key size

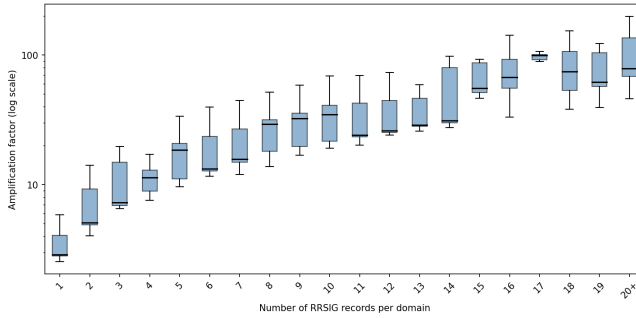


Figure 8: Amplification vs RRSIG record count

dampening the apparent effect of key growth. As key sizes increase, the payload becomes dominated by key material, leading to larger marginal increases in amplification at higher sizes.

RRSIG

Record Count Reflecting the analysis of DNSKEY, we examine the relationship between the records covered by the signature and the amplification of the query. Figure 8 plots the 25-75 percentile amplification for each number of RRSIG records returned. As expected, there is a near linear relationship between the amplification and number of records. On this log scale plot, this indicates an approximately exponential relationship, suggesting that zone complexity — here measured by the number of signed RRsets — contributes significantly to RRSIG amplification.

Type Coverage Now, we examine the record types that each RRSIG covers. As established above, RRSIG records often return coverage for many different record types, leading us to the question of whether the inclusion of certain types is related to amplification. Table 4 shows the difference in median amplification between domains

Type	N (has)	Med (has)	N (not)	Med (not)	Δ
SOA	6,371	26.51	1,619	2.94	23.57
DNSKEY	6,476	26.23	1,514	2.92	23.31
MX	5,374	27.98	2,616	5.00	22.98
TXT	5,884	27.08	2,106	4.23	22.85
NS	6,435	26.25	1,555	3.73	22.52
NSEC3PARAM	4,403	28.19	3,587	12.48	15.71
SPF	95	34.81	7,895	20.81	14.00
A	6,199	24.95	1,791	12.39	12.56
CDS	1,730	29.62	6,260	17.56	12.06
NAPTR	71	31.98	7,919	20.83	11.15
AAAA	1,416	28.68	6,574	19.04	9.63
CAA	1,458	26.60	6,532	19.29	7.31
TYPE65534	549	26.50	7,441	20.06	6.44
CDNSKEY	653	23.40	7,337	19.86	3.55
NSEC	1,651	21.72	6,339	20.61	1.11

Table 4: Amplification differences by presence of record type. Δ denotes the difference in median amplification between domains with and without the given type.

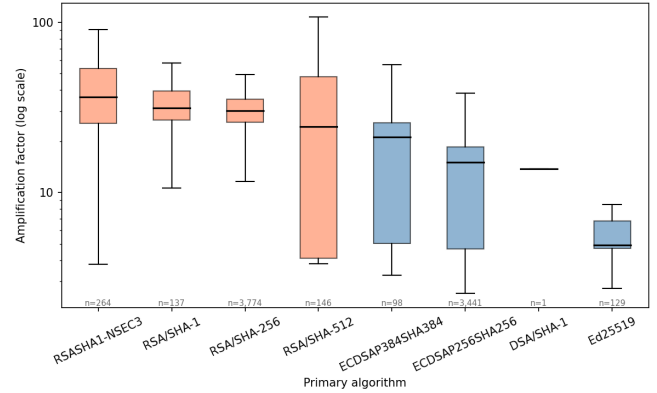


Figure 9: RRSIG Amplification by algorithm (RSA = coral, EC = blue, DSA = green)

that include each record type in their RRSIG response and those that do not. Types such as SOA, NS, DNSKEY, MX, and TXT show the largest gaps, with Δ values exceeding 22. However, these types are not independently responsible for high amplification. They are markers of fully-configured zones that sign many record types simultaneously. The most common combination in our dataset, {A, DNSKEY, MX, NS, NSEC3PARAM, SOA, TXT}, appeared across 1247 domains with a median amplification of 28.1. This suggests that these record types do not individually cause high amplification, instead the breadth of unique records for a domain that drives amplification.

Algorithm As with DNSKEY, we assign each domain a primary algorithm based on the mode across its RRSIG records. Figure 9 shows that RSA domains consistently exhibit higher amplification than EC domains. While the difference in medians is small, the tail end amplification of RSA domains is significantly larger. RSA/SHA-512 domains reach a 95th percentile of 106.0 and 99th percentile of 146.1, whereas the highest 99th

percentile among EC algorithms is 63.1. This mirrors the DNSKEY finding that RSA’s larger signature sizes produce heavier-tailed amplification distributions, and that the shift toward elliptic curve algorithms reduces worst-case amplification risk.

6 Conclusions

This paper presents a large-scale measurement study of DNSSEC amplification across the Tranco top-1M domains. We find that DNSSEC adoption remains low at 6.7%, with signing rates concentrated among higher-ranked domains and .gov domains, where deployment is policy-mandated.

Signed domains produce higher amplification than unsigned ones, with the increase most pronounced in the tail. RRSIG and DNSKEY queries pose the greatest amplification risk, with a maximum amplification of 351.6 and a long tail driven by zone complexity. The number of signed RRsets a zone maintains is the dominant predictor of amplification for both RRSIG and DNSKEY queries.

For DNSKEY, key count, algorithm, and key size all contribute additively to amplification. RSA domains exhibit higher amplification than elliptic curve domains at both the median and tail, and the industry shift toward ECDSA P-256 and Ed25519 has meaningfully reduced worst-case amplification potential.

Together, these findings suggest that while DNSSEC amplification remains a real concern, its risk is concentrated among a specific subset of domains: those that are signed, use RSA algorithms, and maintain complex zone configurations.

To extend the findings of this study, a longitudinal measurement over a longer time period would help determine whether ongoing shifts in zone configurations and algorithm preferences translate into reductions in amplification in practice. As DNSSEC adoption is known to be slow [2], repeated measurements over months or years would also provide a clearer picture of how the amplification landscape evolves as more domains become signed.

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